

Investigating different wires and choosing the most adequate one for a pacemaker lead

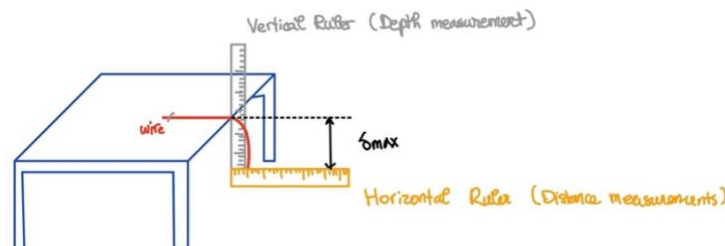
Before Feedback

Method

The aim of this experiment is to find a wire that correspond to the description provided by the researchers in the zoo. The wire must have a diameter of under 4mm^2 , be flexible, and most importantly, be biocompatible. We are provided with 6 wires: red, white, black, brown, copper, and grey.

The first part of the experiment is to measure the diameter of each of these wires. We do so by using a digital caliper. We place the caliper at the edge of the wire and write down the measurements.

The second part is to measure the flexibility. To begin, we place the chosen wire flat on the surface of a table, perpendicular to its edge. We need to take the measurements for three extensions: 8, 13 and 18 cm. Every wire is 30 cm long, so this means that for an 8 cm extension, 22 cm of the wire should be flat on the table. We can fix the wire on the table using some tape, to avoid movements and improper measurements. The diagram below illustrates what the set-up should look like. To continue, we take the measurements using two rulers: one ruler is placed vertically, and measures the depth at which the wire falls, and the other ruler is placed horizontally (parallel to the surface of the table), which is used as a right-angle base for the first ruler. We proceed by writing down the depth measurements, which is δ_{max} .



Feedback

*I assume the wires diameters are given and don't need to be measured (after asking, they do)
Should be written in the past tense*

Decision and Justification

The black wire's cross-sectional area is the only one above 4mm^2 , therefore we can automatically exclude it from our list of possibilities. The remaining wires are all adequate to this criterion.

The higher the flexibility constant, the higher the flexibility of the wire. Before making any decisions, it is important for us to ask ourselves: what type of flexibility are we looking for? Are we looking for flexibility in being able to manually shape the wire, or perhaps flexibility in the movement of the wire itself and how it can bend? Since we are talking about a pacemaker wire, it needs to be flexible in the sense where it can follow the flow of heart pumping and vein movement. If we insert a stiff wire in, it might cause blockages and stop blood from flowing properly through the veins. It can also cause injuries and pain by scratching the veins. Therefore, we are not looking for a wire we can shape. The copper and grey wires are both are very stiff

according to the flexibility constant and our observations, we can therefore eliminate them both from our options.

The last element we should be looking into is biocompatibility. In terms of the insulator used, all their insulators are biocompatible. However, not all are approved by the FDA as pacemaker lead materials. As of today, the only materials from our options that are approved by the FDA to be used as pacemaker insulators are silicone and polyurethane ^[7]. The white and brown wires are made of TPE and PVC respectfully, and neither of these materials are approved.

We can conclude that the best wire that can be used for the pacemaker is the red wire, as it has a cross sectional area smaller than 4 mm^2 , is highly flexible, and is biocompatible. It fits all the required characteristics.

Feedback

Avoid using rhetorical questions

Should be written in the passive form

Structure it more

Try minimizing the use of the personal pronoun "our"

Reference where you got the biocompatibility information

Discussion (before feedback)

There are a couple of possible errors we should consider: firstly, we notice that before every measurement, wires must be pulled and stretched out to be measured correctly. They must be held down to avoid them bending back.

Secondly, stiff wires can be shaped in many ways, we can therefore vary the readings in any way we want. For instance, we can bend the wire down completely and obtain the same measurement we would for an extremely flexible wire, despite the wires' stiff properties in real life. So, the method is not as efficient as we believe, as it can work better for flexible wires than for stiffer ones.

Feedback

Avoid starting a sentence with "So"

Add justifications for why these arguments can be considered as errors

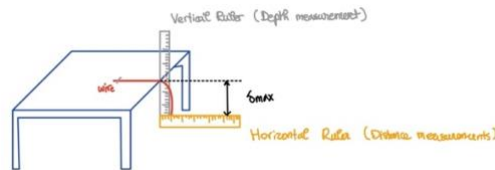
Use passive voice

After Feedback

Method

The wire needed to have a diameter of under 4mm², hence the first part of the experiment was to measure the diameter of each of these wires, using tools such as calipers. The caliper was placed at the edge of the wire, and diameter measurements were written down on a datasheet.

For the second part of this experiment, the flexibility needed to be measured. We were provided with 6 wires: red, white, black, brown, copper, and grey. To begin, the chosen wire had to be placed flat on the surface of a table, perpendicular to its edge. A part of the wire was on the table, and another was in extension. Measurements were taken for three extensions: 8, 13 and 18 cm. Every wire was 30 cm long, so it meant that for an 8 cm extension, 22 cm of the wire needed to be flat on the table. The wire was fixed on the table using some tape, to avoid movements and improper measurements. The diagram below illustrates what the set-up looked like. To continue, two rulers were needed to take the measurements: one ruler was placed vertically and was used to measure the depth at which the wire fell, and the other ruler was placed horizontally (parallel to the surface of the table) and was used as a right-angle base for the first ruler. We proceeded by writing down the depth measurements, and using them to obtain delta max.



Results

Several characteristics were taken into consideration when choosing the most appropriate wire. The first was the cross-sectional area, which had to be smaller than 4 mm². During the lab, we were able to measure the diameter of each wire. Using the $\frac{1}{4}\pi d^2$ formula, the cross-sectional area of every wire was computed.

Secondly, the flexibility constant c had to be calculated. This was done by rewriting the formula $\delta_{\max} = c \times l^4$, such that:

$$c = \frac{\delta_{\max}}{l^4}$$

Since measurements were taken for delta max depending on different lengths l , only one of the extension measurements (say 8 cm for example) was needed to compare the different flexibility constants.

Thirdly, since the wires were intended to be injected in a living organism (elephant), the biocompatibility of the insulator had to be studied. For this reason, the datasheets provided on Moodle were used, and the materials for the insulators of every wire were written down.

Finally, visual observations were added to back up our findings and help us analyze the structures of the wires better.

All the mentioned characteristics can be summarized in the following table. *View appendix for additional data.*

	Black	White	Red	Brown	Grey	Copper
Diameter (in mm)	3.58	1.97	1.89	1.15	0.73	0.30
Cross-sectional area (in mm ²): $\frac{1}{4}\pi d^2$	10.066	3.048	2.805	1.038	0.418	0.071

Flexibility Constant C, for delta max = 0.08m	1,171,875	1,326,497.4	1,074,218.75	56,966.1458	480,143.229	40,690.1042
Insulator material	Silicon/PVC	TPE (thermoplastic elastomer)	Silicon	PVC	ETFE (Ethylene tetrafluoroethylene)	Polyurethane
Biocompatible (FDA approved)	Yes	No	Yes	No	No	Yes
Visual observation	Slightly thick & stiff, still little flexible	Floppy, Flexible	Floppy, Flexible	Stiff, stays in the shape you form	Stiff, not flexible	Very thin, very stiff

Decision and Justification

Firstly, to begin the decision-making process, the first criteria the wire needs to meet is a cross-sectional area lower than 4 mm^2 . Looking at the table of results, the black wire's cross-sectional area is the only one above 4 mm^2 , therefore it can automatically be excluded from the list of possibilities. The remaining wires are all adequate to this criterion.

Secondly, the higher the flexibility constant, the higher the flexibility of the wire. Since it is pacemaker wires that are examined, it is important for them to be flexible such that they can follow the flow of heart pumping and vein movement. By inserting a stiff wire in an organism, blockages may be caused, and blood may stop flowing properly through the veins. Injuries and pain may also be caused due to the wire scratching the veins. Therefore, a wire that can be shaped manually is not efficient, a wire that can bend on its own however is more appropriate. The copper and grey wires are both very stiff according to the flexibility constant and observations, they can therefore be eliminated from the wire options.

Thirdly, the last element that must be studied is the biocompatibility of the wires. In terms of the insulator used, every wire's insulator can be considered as biocompatible ^{[1][2][3][4][5][6]}. However, not all were approved by the FDA as pacemaker lead materials. As of today, the only materials approved by the FDA for the use of pacemaker insulators are silicone and polyurethane ^[7]. The white and brown wires are made of TPE and PVC respectfully, and neither of these materials are approved. They can therefore be excluded from the wire options too.

Thus, it can be concluded that the most suitable wire for the elephant's pacemaker is the red wire, as it has a cross sectional area smaller than 4 mm^2 , is highly flexible, and is biocompatible. The researchers' required characteristics have been met.

Discussion

While this method may be an effective way of measuring the flexibility of several types of wires, there are multiple possible errors that must be considered.

Firstly, it can be noticed that before every measurement, wires must be pulled and stretched out to be measured correctly. They must be held down to avoid them bending back. This can be considered as inaccurate, as the wires will not be that straight in a real organism.

Secondly, stiff wires can be shaped in many ways, therefore, readings may be varied however wanted. For instance, the wire can be bent down completely, and measurements of an extremely flexible wire can be obtained, despite the wires' stiff properties in real life. This leads to misleading information and ambiguities. In fact, the wire can be bent upwards, and negative values can be found. Therefore, the method may not be as efficient as we believe and can be considered more suitable for flexible wires rather than stiff ones.

References:

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[7] (July 18, 2018), U.S. Department of Health and Human Services, Food and Drug Administration, Center for Devices and Radiological Health, *Submission of Research and Marketing Applications for Permanent Pacemaker Leads and for Pacemaker Lead Adaptor 510(k) Submissions - Guidance for Industry*, retrieved on November 3, 2022, from,

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Appendix

Data used to obtain the summarized table in the results section.

Results of flexibility testing						
	Delta max, (mm)					
Length/m	Black	Red	White	Brown	Copper	Grey
0.08	49	55	35	2	1	2
0.08	48	52	49	3	1	1
0.08	47	56	48	2	57	2
0.13	95	109	108	4	15	3
0.13	99	114	107	5	45	3
0.13	100	109	109	4.5	38	3
0.18	148	151	142	10	80	5
0.18	155	148	150	7	55	6
0.18	151	155	153	9	13.5	4

	Average delta max, (mm)					
Length/m	Black	Red	White	Brown	Copper	Grey
0.08	48	54.3333333	44	2.33333333	19.6666667	1.66666667
0.08						
0.08						
0.13	98	110.666667	108	4.5	32.6666667	3
0.13						
0.13						
0.18	151.333333	151.333333	148.333333	8.66666667	49.5	5
0.18						
0.18						

	Flexibility Constant					
Length/m	Black	Red	White	Brown	Copper	Grey
0.08	1171875	1326497.4	1074218.75	56966.1458	480143.229	40690.1042
0.13	343125.241	387474.762	378138.02	15755.7508	114375.08	10503.8339
0.18	144159.935	144159.935	141302.139	8255.85531	47153.6351	4762.99345